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SUBJECT CODE NO: - RNP-8011
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A) Mech/ Civil (NEP) Sem - II
Examination June/July 2025
Basics of Mechanical Engineering

[Time: 3 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

SECTION A**Q1 Attempt any 10 question from following****20**

1. What does the Zeroth Law of Thermodynamics state?
 - a) Energy cannot be created or destroyed
 - b) If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other
 - c) Heat flows from low temperature to high temperature
 - d) Work is done when heat flows from a system
2. Which law of thermodynamics introduces the concept of energy conservation?
 - a) Zeroth Law
 - b) First Law
 - c) Second Law
 - d) Third Law
3. Which type of energy source is considered non-conventional?
 - a) Coal
 - b) Natural gas
 - c) Solar energy
 - d) Nuclear energy
4. Which of the following is NOT a thermodynamic system type?
 - a) Closed system
 - b) Open system
 - c) Isolated system
 - d) Fluid system
5. Which of the following is a key feature of a four-stroke engine?
 - a) It completes a cycle in two strokes
 - b) It completes a cycle in four strokes
 - c) It uses continuous fuel combustion
 - d) It has a high compression ratio
6. The refrigeration cycle is based on the principle of
 - a) Absorption of heat from the environment
 - b) Compression and expansion of gases
 - c) Conversion of work into heat
 - d) None of the above
7. A common property of bearings is to
 - a) Transmit power
 - b) Reduce friction
 - c) Increase load capacity
 - d) Provide mechanical advantage
8. Which type of gear is used to transfer motion between non-parallel shafts?
 - a) Spur gear
 - b) Bevel gear
 - c) Worm gear
 - d) Rack and pinion

9. What is the primary difference between annealing and normalizing?
 - a) Annealing is done at higher temperatures than normalizing
 - b) Annealing involves rapid cooling, whereas normalizing uses slow cooling
 - c) Normalizing is done to relieve internal stresses, while annealing improves ductility
 - d) Annealing increases hardness, whereas normalizing reduces hardness
10. Which of the following is a metal forming process?
 - a) Casting
 - b) Forging
 - c) Welding
 - d) Machining
11. Which machine tool is specifically designed for drilling holes in a workpiece?
 - a) Lathe Machine
 - b) Milling Machine
 - c) Drilling Machine
 - d) Shaper Machine
12. What is the primary function of a shaper machine?
 - a) To remove metal from the workpiece using a rotating tool
 - b) To cut grooves or shapes on a workpiece with a reciprocating tool
 - c) To grind a workpiece to smoothen its surface
 - d) To drill holes in the workpiece

SECTION B

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| Q2 | a) Explain thermodynamics equilibrium in details. | 05 |
| | b) what is pressure? Explain its any one type of pressure measuring device. | 05 |
| Q3 | a) State the various modes of heat transfer and explain them briefly | 05 |
| | b) Write short note on forms of energy. | 05 |
| Q4 | a) Explain working principle of Refrigerator. | 05 |
| | b) Explain 4 stroke SI engine. | 05 |
| Q5 | a) State and explain the selection criteria for Engineering Materials | 05 |
| | b) What is heat treatment process? Explain in detail normalizing heat treatment process. | 05 |
| Q6 | a) Explain with neat sketch the various operations carried out on drilling machine. | 05 |
| | b) Explain CNC in Details. | 05 |
| Q7 | a) Differentiate between flat and v belt drive. | 05 |
| | b) Explain the construction and working of single plate clutch with neat diagram. | 05 |